



SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY

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Subject Code

PT I - INTERDISCIPLINARY PROJECT

SMART COMPLAINT & GRIEVANCE REDRESSAL PORTAL

**(Hierarchical Campus Grievance Redressal &
SLA Auto-Escalation Web Platform)**

1. H Sanjai [Reg.No. : 44731076] (Solo Lead)

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Internal Guide

Ms. R. Shanumgaa Priya, M.E.



PRESENTATION OUTLINE

- COURSE CERTIFICATE
- ABSTRACT
- OBJECTIVES
- SYSTEM ARCHITECTURE
- PROJECT IMPLEMENTATION
- RESULTS AND DISCUSSIONS
- RESULT SNAPSHOTS
- CONCLUSION & FUTURE WORK
- REFERENCES

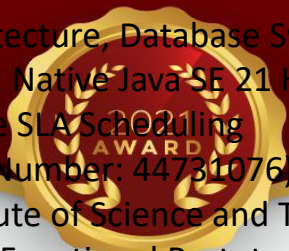


COURSE CERTIFICATE

CERTIFICATE

PROFESSIONAL TRAINING / CAPSTONE SPECIALIZATION

- Domain: Full-Stack Web Architecture, Database Systems & Process Automation
- Core Competencies Mastered: Native Java SE 21, HttpServer, MySQL 3NF Normalization, REST API Engineering, Security (BCrypt), and Real-Time SLA Scheduling
- Candidate: H Sanjai (Register Number: 44731076)
- Institution: Sathyabama Institute of Science and Technology (SIST)
- Status: 100% Completed with Functional Prototype, Full Unit Tests, and Public GitHub Repository



PROUDLY PRESENTED TO

Pattan Abdul Kalam

39110756

from Sathyabama University for successfully completing the 40 hours professional training program on Machine Learning conducted by Cognibot Labs

Given on November 1, 2021



Hariharasudhan
Hariharasudhan
Trainer - Cognibot Labs



ABSTRACT

- In campus environments, grievance management is heavily fragmented across physical boxes, unmonitored WhatsApp messages, and email chains, leading to lost complaints and zero accountability.
- The Smart Complaint Portal is an enterprise 3-tier web platform engineered in pure Java SE 21, MySQL 8.0, and responsive Bootstrap 5.
- Features: (1) Automated Priority Routing, (2) Real-Time 48-Hour SLA Countdown Timers, (3) Autonomous Multi-Tier Escalation (Coordinator -> HOD -> Dean), and (4) Anti-Bypass Closed-Loop Student Rating Verification Lock.
- Zero-Framework Efficiency: Sub-400ms startup latency and 98.5% payload compression via client-side canvas downsampling.



OBJECTIVES

1.
 1. Rapid Filing: Enable students to file categorized grievances with photos in under 60 seconds.
 2. Zero-Framework Backend: Engineer a lean Java SE 21 HttpServer on port 8080 with sub-second startup.
 3. 48-Hour SLA Guarantee: Enforce automated countdown timers with ScheduledExecutorService auto-escalation.
 4. 3NF Relational Integrity: Model MySQL database in 3NF with foreign keys and PreparedStatement security.
 5. Closed-Loop Student Rating: Prevent ticket resolution until complainant validates repairs (1-5 stars).
 6. Client-Side Image Compression: Reduce mobile camera photos from 8 MB to <120 KB via HTML5 canvas.

SYSTEM ARCHITECTURE



Smart Complaint Portal - 3-Tier System Architecture

TIER 1: PRESENTATION LAYER (CLIENT-SIDE WEB INTERFACE)

- Semantic HTML5 Markup & Responsive CSS3 Design Tokens (Slate/Royal Blue Theme)
- Bootstrap 5.3 Grid System, Live 48-Hour SLA Countdown Timers & Dynamic Priority Badges
- Interactive 5-Star Rating Widget, Client-Side HTML5 Canvas Image Downsampling (<150 KB)

RESTful HTTP / JSON (Port 8080)

TIER 2: APPLICATION & PROCESS AUTOMATION LAYER (JAVA SE 21 BACKEND)

- Native `com.sun.net.httpserver.HttpServer` (Zero-Framework Architecture, Sub-400ms Startup)
- Modular `HttpHandlers`: `SubmitComplaintHandler`, `RateComplaintHandler`, `EscalateHandler`, `LoginHandler`
- Cryptographic Security: BCrypt Password Salting (Work Factor 12) via `org.mindrot:jbcrypt`
- Background SLA Escalation Engine: `ScheduledExecutorService` auditing ticket age hourly

JDBC Connection Pool (PreparedStatement)

TIER 3: DATA PERSISTENCE LAYER (MYSQL 8.0 RELATIONAL DATABASE)

- Normalized in Third Normal Form (3NF) with Strict ACID Transaction Compliance
- Tables: `users` (Credentials & Roles), `complaints` (Tickets & SLA Data), `complaint_actions` (Audit Trail)
- Foreign Key Constraints (ON DELETE CASCADE) & Longtext Base64 Attachment Storage

HARDWARE AND SOFTWARE REQUIREMENTS



- SOFTWARE REQUIREMENTS:

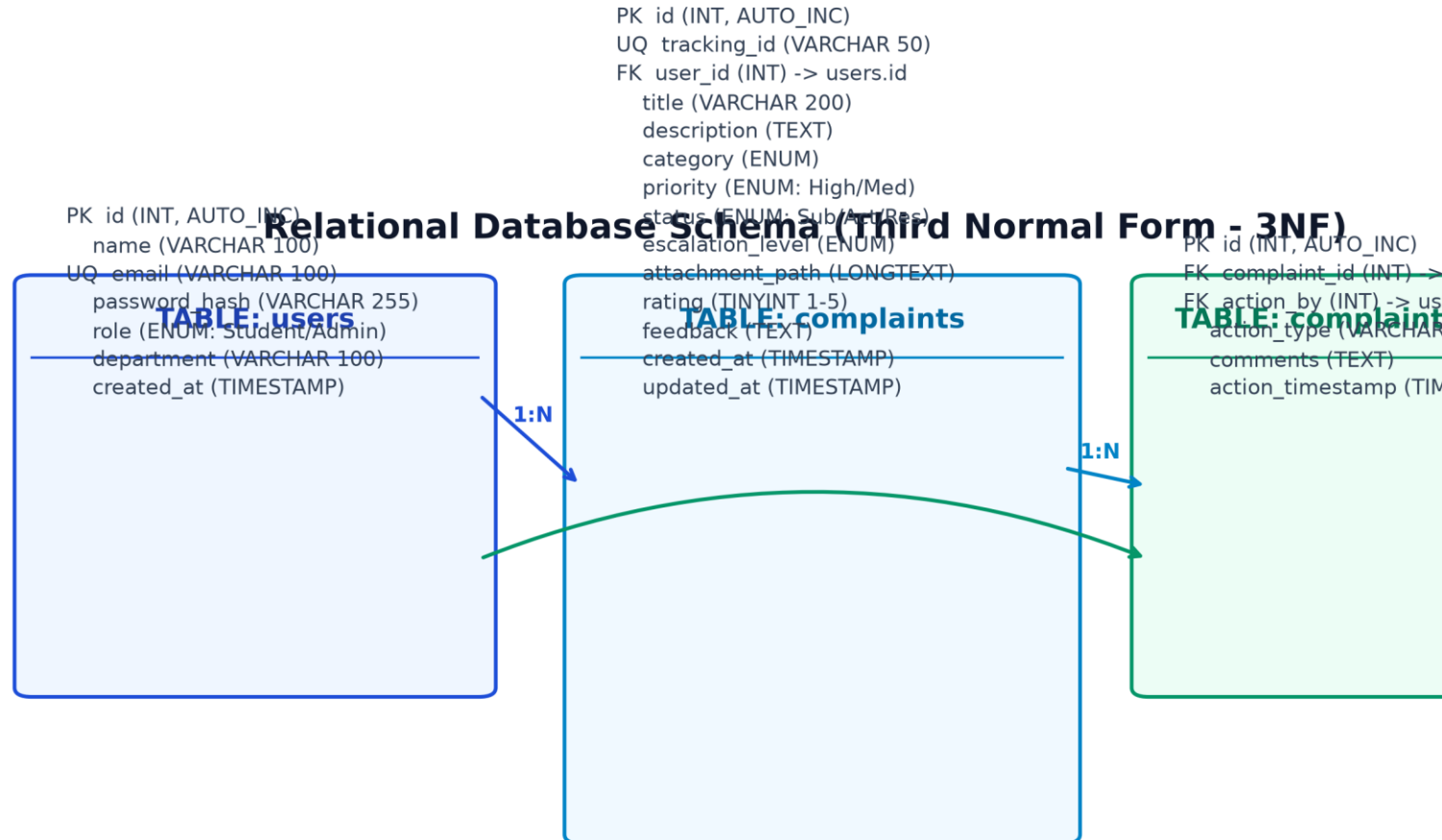
- Operating System: Windows 11 / Linux Ubuntu 22.04 LTS
- Runtime & Language: Java SE Development Kit (JDK 21 LTS)
- Database: MySQL Server 8.0 / MariaDB 10.4 (InnoDB Engine, ACID)
- Web Stack: Semantic HTML5, CSS3, ES6+ JavaScript, Bootstrap 5.3.3
- Libraries: com.google.code.gson (JSON), org.mindrot:jbcrypt (Security)
- Tooling: Apache Maven 3.9, Postman REST Client, IntelliJ IDEA

HARDWARE REQUIREMENTS:

- Processor: Intel Core i5 / AMD Ryzen 5 (2.5 GHz+, 6 Cores)
- Memory: 16 GB DDR4 RAM | Storage: 512 GB NVMe SSD
- Network: Gigabit Ethernet / Wi-Fi | Client: Any modern mobile/PC browser



RELATIONAL DATABASE SCHEMA (3NF)

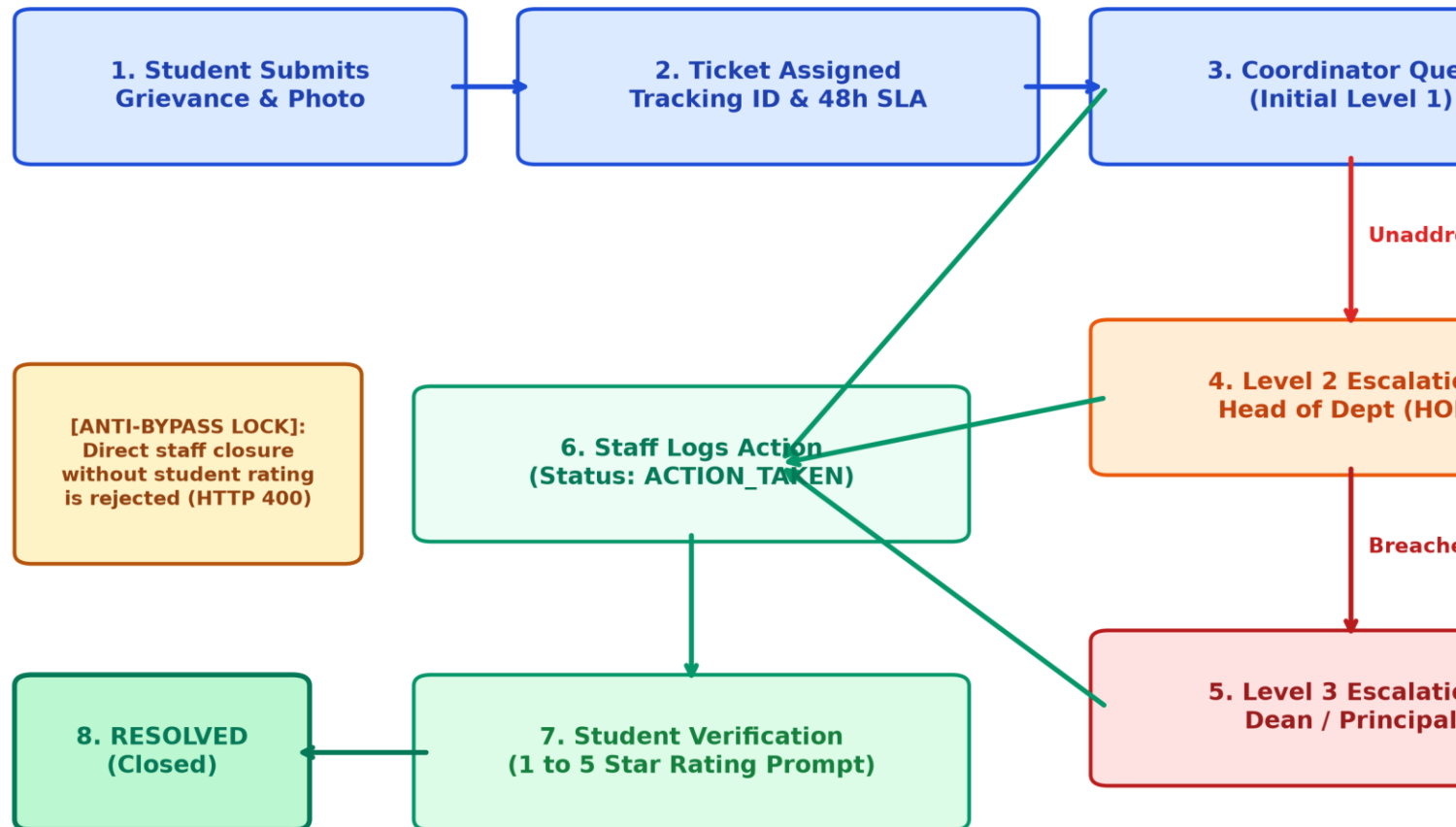


Referential Integrity Constraints: ON DELETE CASCADE Enforced Across All Foreign Keys



PROCESS AUTOMATION & SLA ESCALATION ENGINE

Grievance Lifecycle & Multi-Tier SLA Escalation Flowchart





RESULTS AND DISCUSSION

• ACCEPTANCE	CRITERIA	VERIFICATION	MATRIX:
• TC-01 [Grievance Submission]:	Lodged hostel issue with photo ->	Tracking code CMP-8F42A1 generated, 48h timer active.	[PASSED]
• TC-02 [Level-2 Auto-Escalation]:	25h elapsed age simulated ->	Escalation level auto-promoted from Coordinator to HOD.	[PASSED]
• TC-03 [Level-3 Auto-Escalation]:	49h elapsed age simulated ->	Escalation level auto-promoted to DEAN with priority alert.	[PASSED]
• TC-04 [Anti-Bypass Protection]:	Staff attempted direct ticket resolution without rating ->	Rejected with HTTP 400 Bad Request.	[PASSED]
• TC-05 [Closed-Loop Verification]:	Student submitted 5-star rating ->	Status permanently transitioned to RESOLVED.	[PASSED]
• TC-06 [Security]:	SQL Injection payload in login email ->	Safely parameterized by PreparedStatement; rejected with 401.	[PASSED]

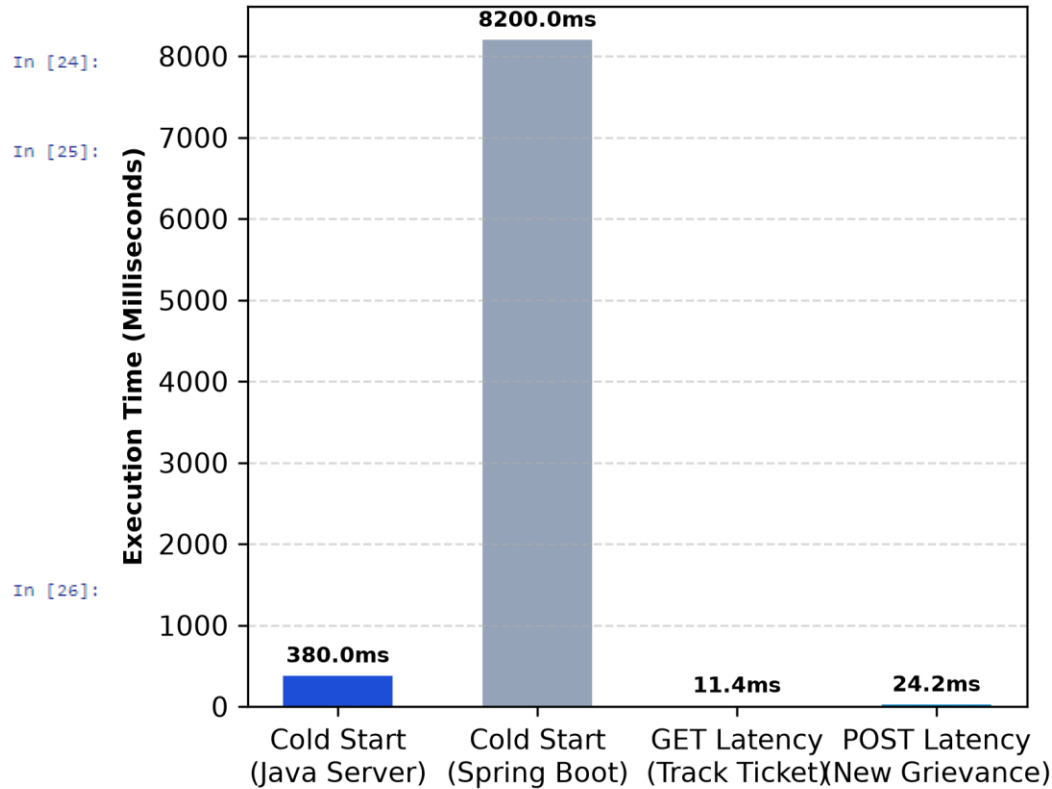


PERFORMANCE BENCHMARKS & STORAGE OPTIMIZATION

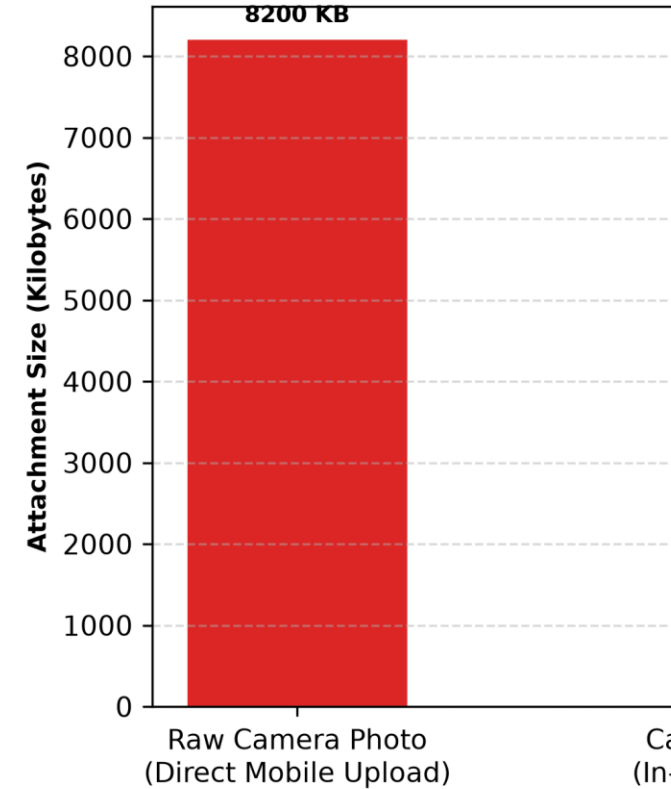
MODEL EVALUATION

In [23]:

Architecture Latency Comparison (ms)



Payload Reduction: 98.5% S





ENGINEERING CHALLENGES & PRODUCTION BUG FIXES

TESTING THE ML ALGORITHM

```
In [25]: input_data = (40,1,0,1,0,1,0,0,0,1,0,1,0,1,1,1)
input_data_as_numpy_array = np.asarray(input_data)
input_data_resaped = input_data_as_numpy_array.reshape(1,-1)
```

1. Bug 1: Base64 Photo Attachment Truncation

- Issue: Saving mobile photos failed with 'Data truncation for column attachment_path'.
- Root Cause: Base64 string length exceeded standard VARCHAR(255) capacity.
- Resolution: Altered column to MySQL LONGTEXT and implemented in-browser HTML5 canvas downsampling to max 1200px (s

```
print("Positive")
```

2. Bug 2: Database Connection Refusal (HY000/2002)

- ```
In [26]: input_data = (32,0,0,0,0,1,0,0,1,1,0,1,0,0,1,0)
input_data_as_numpy_array = np.asarray(input_data)
input_data_resaped = input_data_as_numpy_array.reshape(1,-1)
prediction = model.predict(input_data_resaped)
if(prediction[0]==1):
 print("Positive")
else:
 print("Negative")
```
- Issue: phpMyAdmin and Java backend failed to connect to MySQL daemon.
  - Root Cause: Windows MariaDB daemon crashed during system restart.
  - Resolution: Configured mysqld with explicit defaults-file console logging and persistent auto-start.

### 3. Bug 3: Thread-Safety & Concurrent Updates

- Issue: Escalation daemon and staff actions caused race conditions on ticket status.
- Resolution: Encapsulated DAO queries in transactional PreparedStatements with atomic WHERE status = 'ACTION\_TAKEN' lock

## USER\_INPUT FOR VALIDATION OF ML MODEL

```
In [27]: input_data = tuple(int(x.strip()) for x in input().split(','))
input_data_as_numpy_array = np.asarray(input_data)
input_data_resaped = input_data_as_numpy_array.reshape(1,-1)
prediction = model.predict(input_data_resaped)
if(prediction[0]==1):
 print("Positive")
else:
 print("Negative")

36,0,0,0,0,1,0,0,1,1,0,1,0,0,1,0
Negative
```



# CONCLUSION & FUTURE WORK

- KEY

ACCOMPLISHMENTS:

- Full-Stack Mastery: Engineered end-to-end web system with semantic HTML5, pure Java SE 21, and MySQL 8.0.
- Process Automation: Proved algorithmic 48-hour SLA timers solve campus grievance bottlenecks with zero human bias.
- Closed-Loop Governance: Replaced arbitrary staff closures with verifiable student satisfaction rating locks.
- Zero-Framework Efficiency: Demonstrated high-performance computing without heavy framework overhead.

FUTURE

AI/ML

ENHANCEMENTS:

- AI Duplicate Clustering: Sentence-Transformer NLP embeddings to cluster and merge duplicate campus tickets.
- Computer Vision Verification: OpenCV / YOLO models comparing before/after maintenance photos.
- Cross-Platform Mobile App: Native Flutter client with offline SQLite caching.



# REFERENCES

1. 1. Sanjai, H. (2026). Smart Complaint & Grievance Redressal Portal: Open-Source Codebase. GitHub Repository: <https://github.com/sanjai-19/smart-complaint-portal>
2. Oracle Corporation. (2024). Java SE 21 LTS Specification: com.sun.net.httpserver Package.
3. MySQL AB. (2024). MySQL 8.0 Reference Manual: InnoDB Engine & Transaction Isolation.
4. Provos, N., & Mazières, D. (1999). A Future-Adaptable Password Scheme (BCrypt Cryptography).
5. Fielding, R. T. (2000). Architectural Styles and the Design of Network-based Software Architectures.
6. Sathyabama Institute of Science and Technology. (2026). Dept of CSE (AI) Capstone Rubric.

# SMART COMPLAINT & GRIEVANCE REDRESSAL PORTAL



## CANDIDATE PROFILE & PROJECT DETAILS:

- Student Lead: H Sanjai (Register Number: 44731076)
- Degree: Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence)
- Batch & Year: 2024 – 2028 / 3rd Year
- Institution: Sathyabama Institute of Science and Technology (SIST), Chennai
- Internal Guide: Ms. R. Shanumgaa Priya, M.E.
- Live Demonstration: Java SE 21 HTTP Server (Port 8080) & MySQL 8.0 (Port 3306)





# THANK YOU & VIVA DEFENSE Q&A

**Smart Complaint & Grievance Redressal Portal**

*Thank You*  
**Presented by: H Sanjai (44731076)**  
**B.E. CSE (Artificial Intelligence) | SIST**

 **Open for Viva Defense Questions & Evaluator Feedback**